A3-PS019-TC374

Personal-Data Discovery and Privacy Evidence Inventory Engine

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Problem Description

Frameworks like the DPDP Act 2023 and GDPR require organizations to locate stored PII and map it to purposes and retention timelines.

- **Core issue:** unindexed tables, legacy CSV exports and spreadsheets accumulate "dark" personal data with no audit trail or metadata.
- **Manual PII inventories** are slow, error-prone and go stale quickly.
- **Over-automation is risky:** tools that auto-declare lawful basis make legal inferences that create liability.
- **Objective:** a technical discovery engine that inspects CSVs and SQL databases, maps candidate fields across 5 predefined PII classes to declared purpose / retention metadata, and reports documentation gaps.
- **Boundary:** no unauthorized legal decisions - evidence only, for human review.

Background

Context needed to understand the problem and the proposed optimization.

- **Regulatory context:** the DPDP Act and global regulations mandate data minimization, clear purpose specification and strict retention limits [1].
- **Data growth vs. governance:** distributed architectures spread duplicate PII fields across staging tables, backup CSVs and production SQL instances.
- **Technical need:** an automated, rule-driven scanner acting as an objective evidence collector.
- **Boundary definition:** the system supplies technical evidence and quality metrics for review by a human Data Protection Officer (DPO), instead of unverified AI legal decision-making.

Literature & Standard Review

Most relevant existing frameworks and standards.

- **MeitY / DPDP Act (India) [1]:** obligations for Data Fiduciaries on purpose limitation, storage limitation and data inventory.
- **ISO/IEC 27701 & 27018 [2]:** controls for PII protection and requirements for maintaining processing-record inventories.
- **NIST Privacy Framework [3]:** inventory management via data-flow mapping and flagging of missing metadata records.
- **OWASP Data Protection Guide [4]:** developer-level practices for handling personal data.

Key takeaway for PS-019: technical scanning must stay deterministic, auditable, and separated into distinct metadata-detection and human-verification tiers.

Proposed Solution

Dual-engine scanner for SQL databases (MySQL / PostgreSQL / SQLite) and flat CSV files.

Direct Identifiers	Contact Data	Financial Identifiers	Biographic & Demographics	Network & Telemetry
Full name, Aadhaar / National ID, Passport no.	Email, phone number, physical address	Bank account no., UPI ID, credit card no.	Date of birth, gender, nationality	IP address, device ID, MAC address

Core functionality

- **Rule engine:** regex pattern matching + column-header semantics + data-type inspection to identify candidate fields.
- **Metadata binding:** links discovered fields to the organizational data catalog (purpose_registry.json).
- **Gap reporting:** flags fields lacking an explicit business purpose or retention policy tag.

Optimization Proposed

Before vs. after optimization.

Operational metric	Manual / legacy approach	Our optimized discovery engine
Data source scanning	Manual inspection of spreadsheets	Automated parsing of SQL schemas and CSV files
PII identification	Static keyword search on column names	Hybrid rule engine: header semantics + regex + data profiling
Compliance risk	Over-reliant on automated legal assumptions	Strict separation of technical evidence from legal interpretation
Audit preparation	Days / weeks of manual report compiling	Real-time precision / recall score and missing-metadata gap report
Verification loop	No ground-truth verification	Evaluation against a manually labeled test set

Solution Architecture & Design

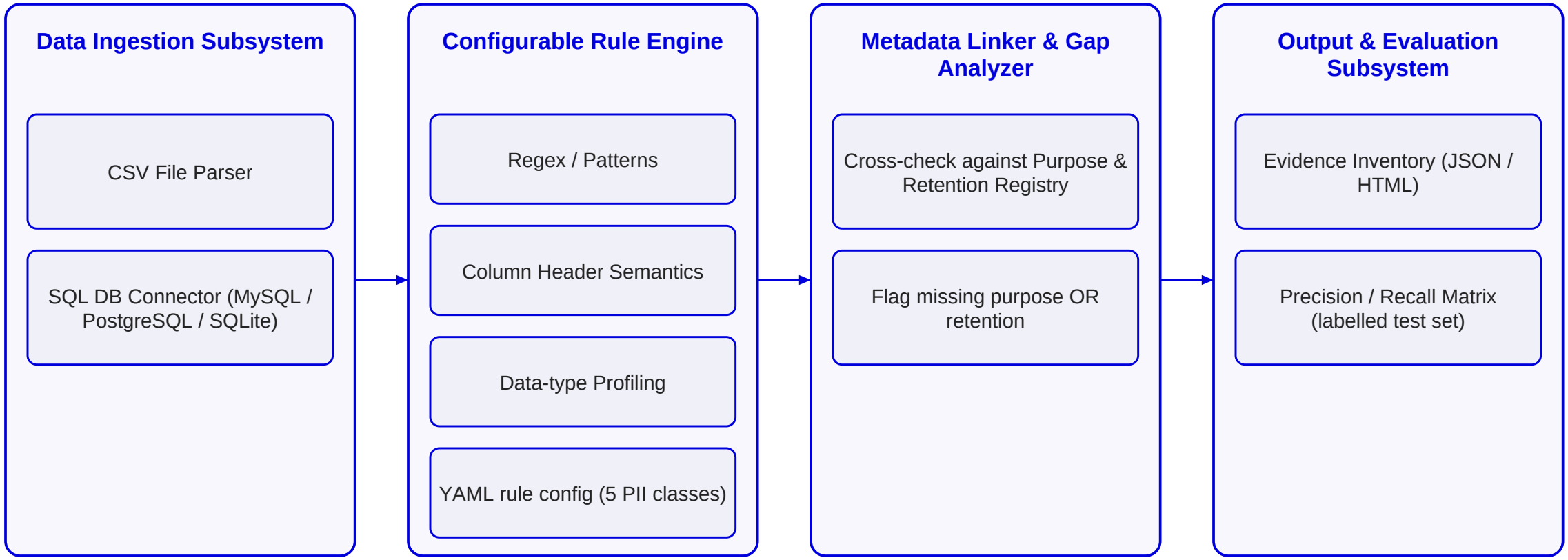

Figure: System Architecture UML Diagram

Data Flow Diagram

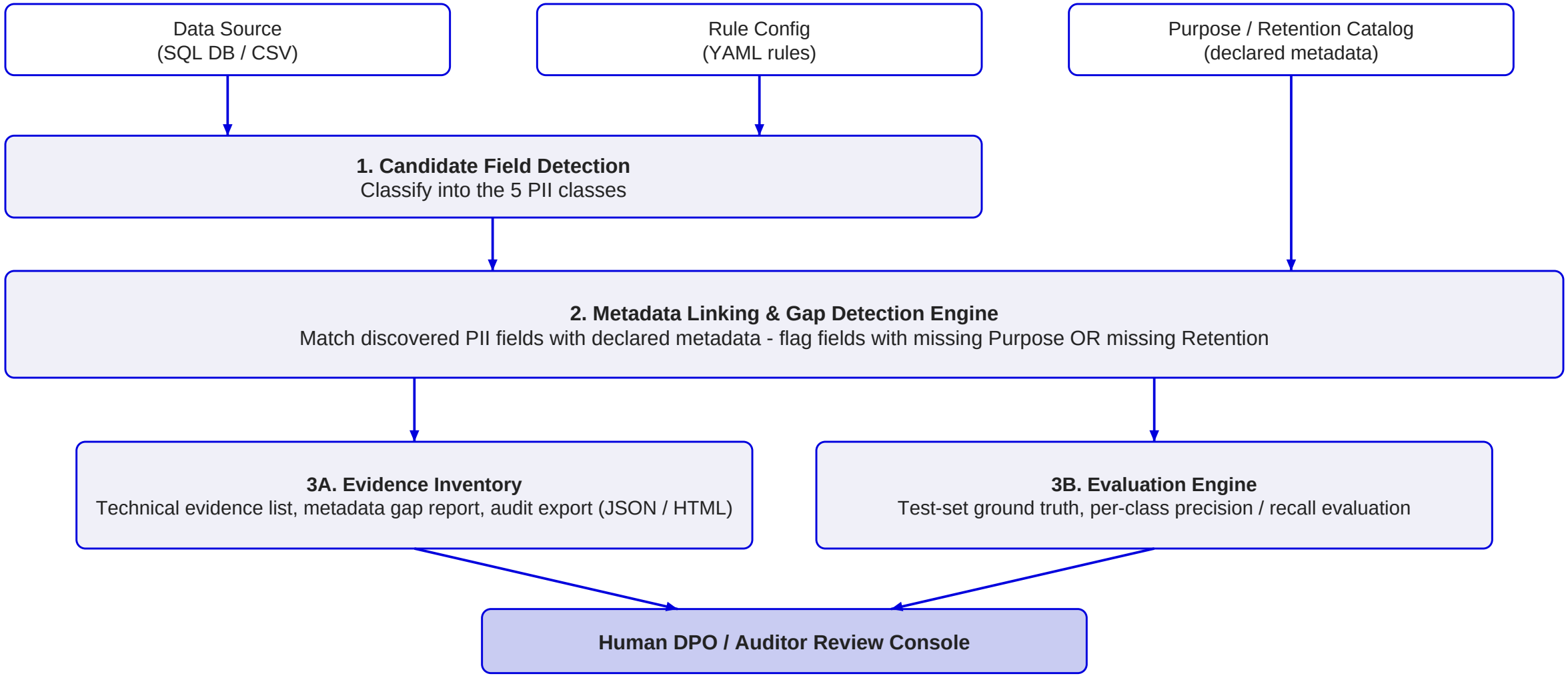

Figure: Data Flow Diagram (DFD)

Deployment Timeline

Milestones and methodology over three days.

Phase 1 - Day 1
Architecture & Rule Engine Development

- Implement CSV and SQLite / MySQL connection modules.
- Define rule patterns (regex + column names) for the 5 target PII classes.

Phase 2 - Day 2
Metadata Linker & Evaluation Module

- Build the purpose and retention mapping module (purpose_registry).
- Build the missing-metadata and evidence-gap reporting module.
- Create a manually labeled benchmark test set for per-class precision and recall.

Phase 3 - Day 3
Integration, Testing & Submission

- Run end-to-end scanner tests on synthetic SQL schemas and CSV files.
- Generate evaluation metrics, complete documentation, upload final PDF to GitHub.

Bibliography

[1] MeitY (Ministry of Electronics and Information Technology): Digital Personal Data Protection Act (DPDP Act) & Data Protection Framework Guidelines, Government of India.

[2] ISO/IEC 27701:2019: Security techniques - Extension to ISO/IEC 27001 and ISO/IEC 27002 for privacy information management - Requirements and guidelines.

[3] National Institute of Standards and Technology (NIST): NIST Privacy Framework: A Tool for Improving Privacy Through Enterprise Risk Management, Version 1.0.

[4] OWASP Foundation: OWASP Data Protection and Privacy Guide for Developers.

Thank You for Your Attention!

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